

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA1008	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	B. Pushparaj	Reg. No.:	192421136
Date:	21.8.2026	Duration:	60 minutes

Code of Conduct

I B. Pushparaj(192421136) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____



Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, design and document comprehensive test cases to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the functional and non-functional requirements to be tested.
2. Identify the test scenarios for the assigned problem statement.
3. Prepare detailed test cases covering normal, boundary, invalid, and exceptional conditions.
4. Include Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status.
5. Apply appropriate test design techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.

6. Ensure that the test cases provide adequate requirement and functional coverage for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis,	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied

	Decision Table, and State Transition			
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

01. Problem Overview

The **Accessible Learning Platform** is a web-based learning system designed to make online education easier for students with different accessibility needs. The platform provides **screen-reader support, voice navigation, audio learning materials, accessible assessments, course access, and progress tracking**.

Objectives

- Provide accessible learning content for all students.
- Support screen readers and keyboard navigation.
- Provide audio-based learning resources.
- Allow students to take accessible online assessments.
- Help teachers upload and manage learning materials.
- Allow administrators to manage users and courses.

Key Functionalities

ID	Functional Requirement
FR01	User registration and login
FR02	Student/teacher role management
FR03	Course and lesson access
FR04	Screen-reader friendly interface
FR05	Voice navigation
FR06	Audio learning materials
FR07	Accessible online assessments
FR08	Automatic assessment result display
FR09	Student progress tracking
FR10	Teacher content management
FR11	Admin user management
FR12	Logout and session management

2. Functional and Non-Functional Requirements

Functional Requirements

1. Users should be able to register and log in.
2. Students should access available courses.
3. Students should listen to audio learning materials.
4. The system should support keyboard navigation.
5. Voice commands should navigate supported pages.
6. Students should attend online assessments.
7. The system should calculate and display results.
8. Students should view their learning progress.
9. Teachers should upload learning materials.

10. Administrators should manage users and courses.

Non-Functional Requirements

Requirement	Description
Accessibility	Interface should support screen readers and keyboard navigation
Usability	Interface should be simple and easy to understand
Performance	Pages should load within an acceptable time
Security	User passwords and sessions should be protected
Reliability	Assessment results should be stored correctly
Compatibility	System should work on common browsers
Availability	Learning resources should be available when required

3. Test Scenarios

Scenario ID	Test Scenario
TS01	User registration
TS02	User login
TS03	Invalid login
TS04	Course access
TS05	Screen-reader navigation
TS06	Keyboard navigation
TS07	Voice navigation
TS08	Audio learning material
TS09	Online assessment
TS10	Assessment submission
TS11	Result calculation
TS12	Progress tracking
TS13	Teacher content upload
TS14	Admin user management
TS15	Logout/session management

4. Detailed Test Cases

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC01	Valid Registration	Open registration → enter details → Submit	Valid name, email, password	Account created successfully	As Expected	Pass
TC02	Invalid Email	Enter invalid email → Submit	student@	Validation message displayed	As Expected	Pass
TC03	Valid Login	Enter username and password → Login	Valid credentials	Dashboard displayed	As Expected	Pass

TC04	Invalid Login	Enter incorrect password	Wrong password	Error message displayed	As Expected	Pass
TC05	Course Access	Login → Select course	Available course	Course content displayed	As Expected	Pass
TC06	Screen Reader	Enable screen reader → navigate page	Page content	Text/buttons are readable	As Expected	Pass
TC07	Keyboard Navigation	Use Tab/Enter keys	Keyboard only	All important controls accessible	As Expected	Pass
TC08	Voice Navigation	Enable voice navigation → give command	“Open Courses”	Courses page opens	As Expected	Pass
TC09	Audio Lesson	Open lesson → Play audio	Audio file	Audio starts and plays correctly	As Expected	Pass
TC10	Start Assessment	Open assessment → answer questions	Valid answers	Questions displayed correctly	As Expected	Pass
TC11	Submit Assessment	Answer questions → Submit	Completed assessment	Assessment submitted successfully	As Expected	Pass
TC12	Result Calculation	Submit assessment	8 correct / 10	Correct score displayed	As Expected	Pass
TC13	Progress Tracking	Complete lesson → View progress	Completed lesson	Progress updated	As Expected	Pass
TC14	Teacher Upload	Teacher login → Upload material	PDF/audio file	Material uploaded successfully	As Expected	Pass
TC15	Invalid Upload	Upload unsupported file	.exe file	File rejected with message	As Expected	Pass
TC16	Admin Management	Admin → Users → Add user	Valid user details	User created	As Expected	Pass
TC17	Logout	Click Logout	Logged-in user	User returned to login page	As Expected	Pass
TC18	Session Security	Logout → press Back	Previous dashboard	Protected page should not be accessible	As Expected	Pass

5. Boundary Value Analysis

For an assessment containing **10 questions**, boundary values can be tested.

Test Case	Input	Expected Result
BVA01	0 answers	Score = 0
BVA02	1 correct answer	Score calculated correctly
BVA03	9 correct answers	Correct score displayed
BVA04	10 correct answers	Score = 100%
BVA05	More than 10 answers	Extra input rejected/not allowed

6. Equivalence Partitioning

Login Password

Partition	Test Data	Expected Result
Valid	Correct password	Login successful
Invalid	Wrong password	Login rejected
Empty	Blank password	Validation message
Invalid format	Very short password	Validation message

User Email

Partition	Test Data	Expected Result
Valid	student@gmail.com	Accepted
Invalid	student@	Rejected
Empty	Blank	Required-field message

7. Decision Table Testing

Login Decision Table

Condition	Case 1	Case 2	Case 3	Case 4
Valid Username	Yes	Yes	No	No
Valid Password	Yes	No	Yes	No
Expected Result	Login	Error	Error	Error

8. State Transition Testing

Student Assessment

Not Started



Assessment Opened



Questions Answered



Assessment Submitted



Result Generated



Progress Updated

Test conditions include:

- Student starts assessment.
- Student answers questions.
- Student submits assessment.
- System calculates result.
- Progress is updated.

9. Traceability Matrix

Requirement	Test Cases	Coverage
FR01 Registration	TC01, TC02	Covered
FR02 Login	TC03, TC04	Covered
FR03 Course Access	TC05	Covered
FR04 Screen Reader	TC06	Covered
FR05 Keyboard Navigation	TC07	Covered
FR06 Voice Navigation	TC08	Covered
FR07 Audio Learning	TC09	Covered
FR08 Assessment	TC10, TC11	Covered
FR09 Result Calculation	TC12	Covered
FR10 Progress Tracking	TC13	Covered
FR11 Content Management	TC14, TC15	Covered
FR12 Admin Management	TC16	Covered
FR13 Logout/Security	TC17, TC18	Covered

Overall test coverage: 100% of the identified functional requirements.

10. Non-Functional Testing

Test ID	Test Type	Test	Expected Result
NFT01	Accessibility	Test with keyboard	All controls accessible
NFT02	Accessibility	Test with screen reader	Text and controls readable
NFT03	Performance	Open dashboard	Page loads within acceptable time
NFT04	Security	Access dashboard after logout	Access denied
NFT05	Compatibility	Test Chrome/Edge/Firefox	Application works correctly
NFT06	Usability	Test navigation	Simple and understandable interface
NFT07	Reliability	Submit assessment	Result stored correctly

11. Test Execution Result

Test Type	Total Tests	Passed	Failed
Functional Testing	18	18	0
Boundary Testing	5	5	0
Equivalence Testing	7	7	0
Decision Table	4	4	0
State Transition	5	5	0
Non-Functional Testing	7	7	0
Total	46	46	0

Result: The proposed Accessible Learning Platform satisfies the identified functional and non-functional requirements based on the designed test cases.

12. Conclusion

The **Accessible Learning Platform** has been evaluated using functional, non-functional, boundary value, equivalence partitioning, decision table, and state transition testing. The test cases cover normal, invalid, boundary, and exceptional conditions. The designed test suite provides good requirement traceability and helps ensure that the platform is **accessible, reliable, secure, and easy to use**.

